

A  
SEMINAR REPORT

ON  
“PROSTHETIC ARM”

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YEAR 2024-25



## CERTIFICATE

This is to certify that the BE project titled,

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## ABSTRACT

Physical impairment can limit the physical function. Such individual then can be called as a handicap. In cases of individuals with loss of limbs and next to nothing residual capacity, it is very difficult for them to associate with daily activities as well as employment, education, independent living, etc. This project will depict the information relating the substitution of the upper limb of an amputee by Myoelectric Prosthetic Arm. The objective of this project was to redesign a prosthetic arm that would be affordable for common people and be best fit for ages from eighteen to fifty. The advances in science and technology have led to development of externally powered prosthesis that interface directly with the neuromuscular system and recreate some of a normal hand's sophisticated proprioceptive control. This device known as Myoelectric

Arm is based on biological electronic sensors, is battery operated, controlled by microprocessor and driven by motors. It then translates this muscle activity as triggered by the user into information via microcontroller, which control the artificial limbs movements via electric motors. The end result is that the artificial limb moves much like a natural limb, according the mental stimulus of the user. The Myoelectric artificial limb does not require any unwieldy straps or harnesses to function. Instead, it is custom made to fit and attach to the remaining limb. Myoelectric hand/arm component perform better than conventional prosthesis in terms of function, weight, comfort and cosmetics. The final design includes a robust and strong design, which can easily be manipulated by the user. It contains cheap, light, but also strong materials.

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## Acknowledgment

With an immense pleasure and satisfaction, I am presenting this seminar report as part of curriculum of Final Year Electronics & Telecommunication Engineering – 2019 course of Savitribai Phule Pune University, Pune in the academic year 2024-25. I wish to express my sincere gratitude towards all those who have extended their support right from its commencement to its completion. Also, I would like to extend my thanks to our Project Guide Prof.A.P.Yadav for their valuable suggestions and feedback throughout Project work. I acknowledge with deep sense of gratitude and most sincere appreciation, the valuable guidance and unfailing encouragement rendered to me by “Prof. Dr.P.V.Jadhav”, Head of Department and other staff for their proficient and enthusiastic guidance, useful encouragement and immense help.

I also wish to extend my sincere thanks to our project Co-ordinator Prof.S.A.Dhole for their insightful comments and constructive suggestions to improve the quality of this training and valuable suggestions which proved extremely useful and helpful in completion of this industrial training.

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## **List of Abbreviations**

1.DOF: Degree Of Freedom

2.EMG: Electro Myography

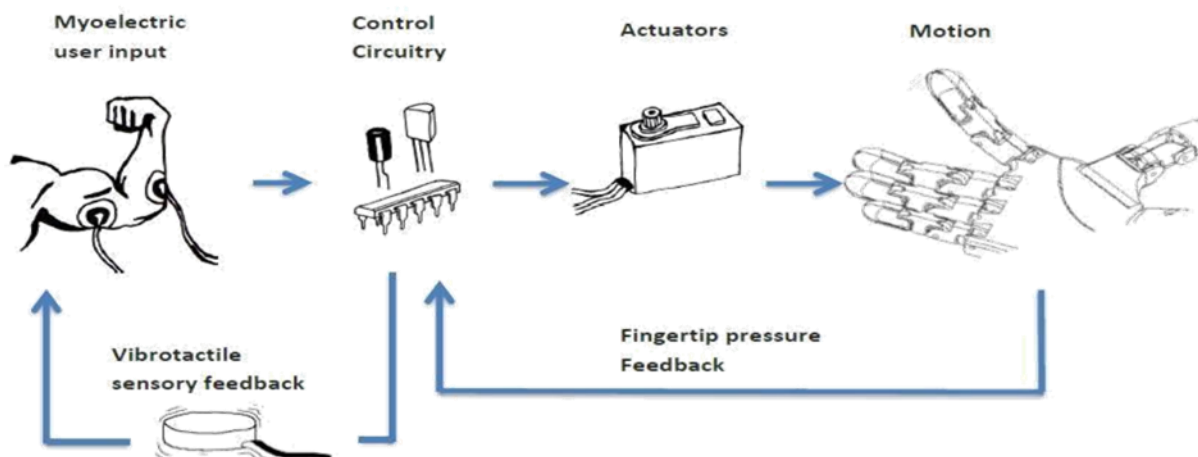
## Introduction

It could be argued that the most valuable possession to any human being is their body. Replacing a missing human limb, especially a hand, is a challenging task which makes one truly appreciate the complexity of the human body. For centuries innovators have been trying to replace lost limbs with manmade devices. Several prosthetic devices have been discovered from ancient civilizations around the world demonstrating the ongoing progress of prosthetic technology.

Until recent times the design of prosthetic limbs has progressed relatively slowly. Early innovations such as the wooden leg can be thought of as simple prosthetic devices. History shows that for a long- time prostheses have remained passive devices that offer little in terms of control and movement.

Over time materials improved and designs started incorporating hinges and pulley systems. This led to simple mechanical body powered devices such as metal hooks which can open and close as a user bends their elbow for example.

Recent times however have given way to enormous advancements in prosthetic devices. Focus is not only on the physical aspects of a device but also the control and biofeedback systems. Slowly we are approaching an advanced trans-human integration between machine and body. Perhaps sometime in the future prosthetic devices will be faster, stronger and maybe even healthier than our biological limbs. Throughout the course of this thesis we will explore myoelectric prosthetic arms. It is aimed to design a device which mimics the function of the human arm as best as possible and can be controlled to some extent by muscular contractions.





## Literature Survey

### 2.1 The human hand

The Human hand comprises at least 27 bones (depending on the individual), more than 30 individual muscles and over 100 named ligaments, nerves, and arteries. As shown in Fig: 2.1. Prostheses aim to replicate the functions of the human body and return functionality to persons with missing extremities. No current prosthetics can match the dexterity, flexibility, and fluidity of the human hand. Before any further discussion, let us briefly explain the meaning of a “degree of freedom” (DOF) for a reader with a non-engineering background.

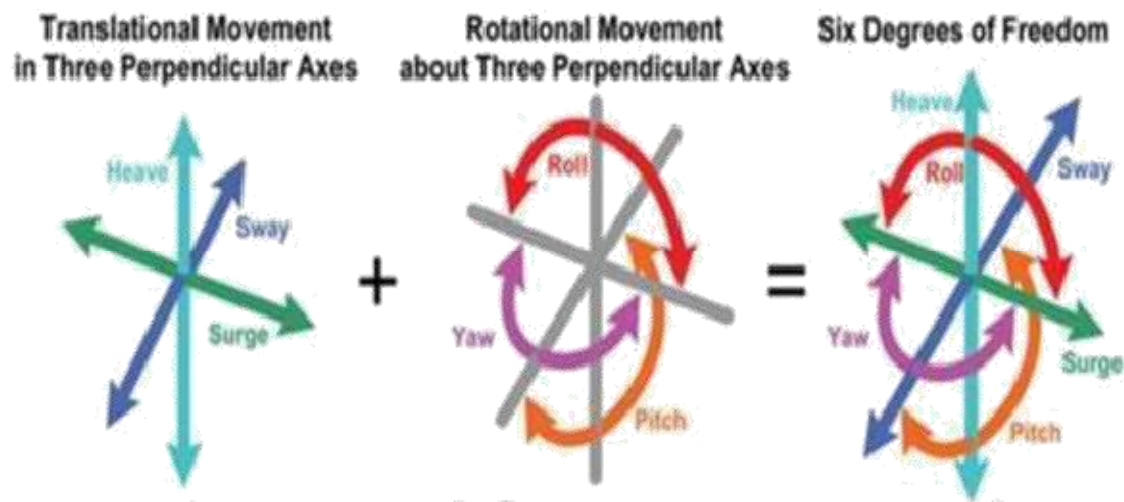


Figure 2.1 – Major Bones in the human hand.

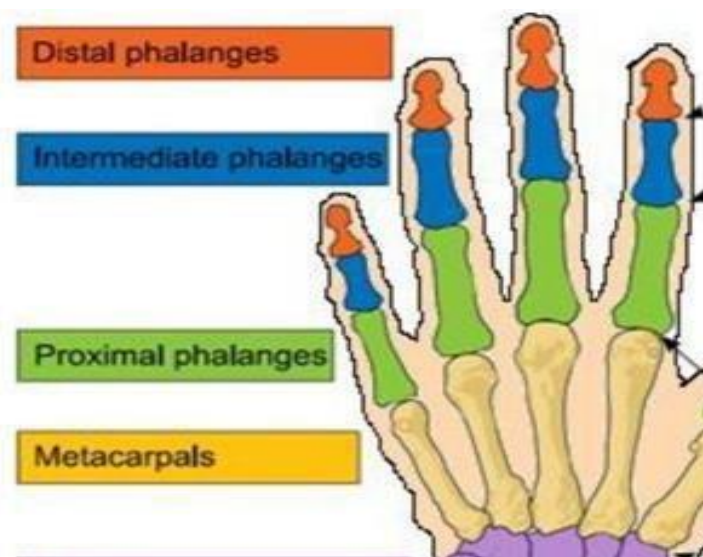


Figure 2.2 – Degrees of freedom at a single point.

Looking at the fig 2.2, imagine a point in space. From this point we can translate (move) along 3 different axes, i.e. we can move forward/backward, up/down and left and right. At the same point we can also rotate around 3 different axes. The human neck for example has 3 degrees of rotational freedom – we can look left/right, up/down and tilt our head sideways. So in total a single point can have a maximum of 6 degrees of freedom (3 translational, 3 rotational).

The human finger in total has 4 degrees of freedom as shown in fig 2.3. Three of these are the rotations of each joint (DIP, PIP, MCP) which combine to control flexion and extension of the finger.

The knuckle (MCP joint) also allows for abduction/adduction (wiggling the finger from side to side).



Figure 2.3 – Depiction of the degrees of freedom in a human finger

In the thumb the lower CMC joint also allows for abduction/adduction – which gives 5 DOFs in the thumb. Finger and all joints in the human body are actuated (moved) via contraction of muscles and tendons.

## 2.2 Capabilities of Prosthetic Hands

The vast majority of commercial prosthetic fingers are actuated through a joint linkage as shown in fig: 2.4 systems powered by DC electric motors. Kinematic models for various prosthetic fingers are shown below. Each finger incorporates its own mechanism to mechanically couple joints together. Rotating the metacarpal joint (knuckle) simultaneously rotates the higher phalange joint.

The problem with this type of design is that there is no control over individual finger joints. All joints in the finger are controlled through a single actuator which means the entire finger has only a single degree of freedom – these fingers can only open/close in a single way. In reality a human finger has control over individual joints so is capable of flexing in a variety of ways, shown in fig 2.4

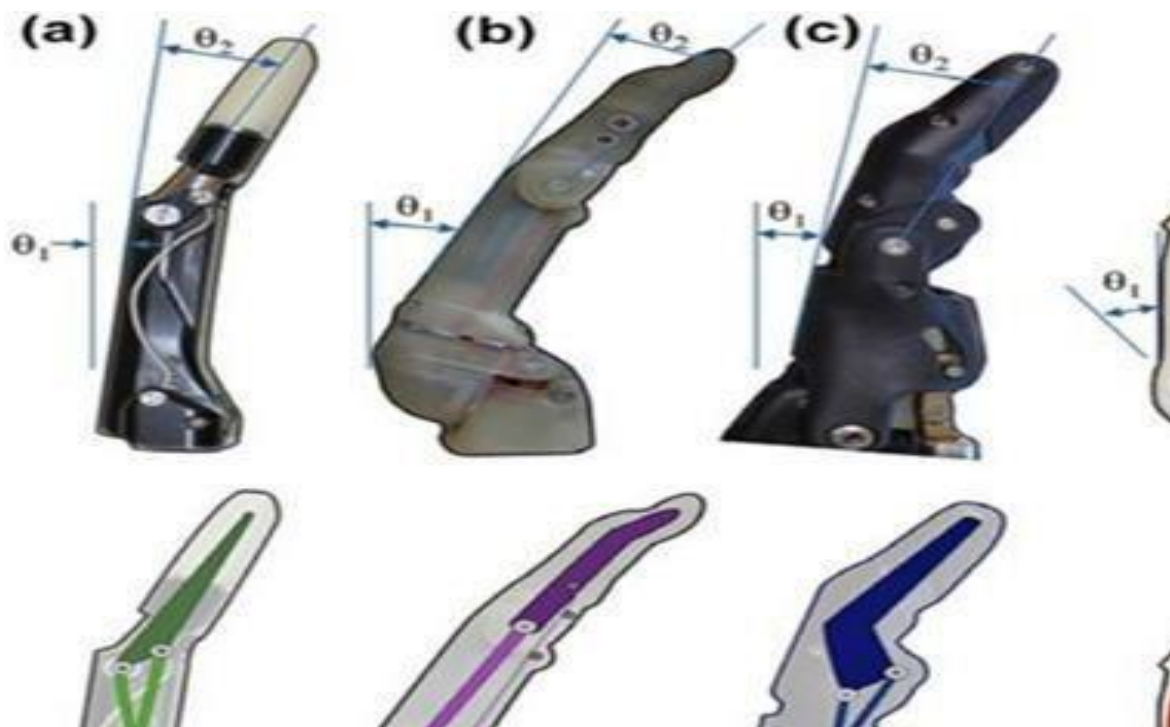


Figure 2.4 – Commercial finger images (top) kinematic models of finger joint coupling mechanism (bottom)

Although commercial prosthetic fingers may have the same number of joints as a human finger, they have fewer degrees of freedom. Usually 1 or 2 compared to



Figure 2.5 – Various flexion arrangements of the human fingers – not possible with modern prosthetic hand

Dexterity arises from the numerous degrees of freedom of the human hand. The fine motor control a person has over their individual finger joints allows for a vast array of intricate tasks to be achieved. In contrast, commercial prostheses are limited to simple tasks partially due to the lack of fine control in the fingers. For example, trying to knit, sew or play a musical instrument like a guitar with a modern commercial prosthetic device would be extremely difficult if not impossible.

Another critical design point in commercial prosthesis is durability. The average user will wear a myoelectric prosthetic hand in excess of 8 hours per day. Therefore, prosthetic arms for commercial use must be robust, lightweight and packaged into a closed system that can be attached to an amputee. Mechanical complexity determines the degrees of freedom in the system; however, there is usually a trade-off because increasing complexity can lead to an increase of the size of the device and also reduce robustness and durability.

### The Bebionic 3

The Bebionic 3 is a world leading commercial myoelectric arm. Like others of its kind, the Bebionic 3 uses a predefined grip system. A user can select from 14 different grip patterns using muscle activity around their upper forearm. The user does not essentially have control of individual finger movements, rather they can select a grip pattern and then use muscle activity to activate the movements of that specific grip. Four of the fourteen grips of the Bebionic 3 are shown in below figure 2.6.

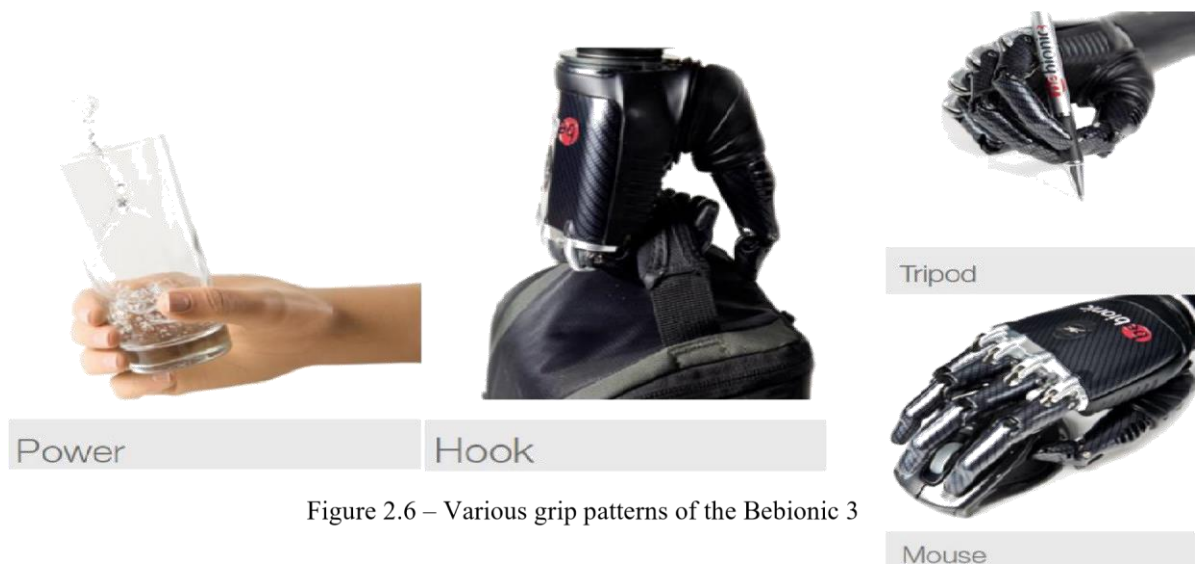


Figure 2.6 – Various grip patterns of the Bebionic 3



The problem with a predefined grip system is that the user cannot finely control finger positions in order to grip a specific object or complete a task. Rather, a user must choose a grip pattern that best suits the job at hand and then actuator that grip pattern.

Furthermore, the user must cycle through a number of grip patterns before they get to their desired choice. For example, unzipping a bag, picking up a heavy object, placing it in that bag and then zipping the bag up could require a number of grip changes. As a result certain simple tasks like this could actually take quite some time to complete and can become tedious and frustrating.

The thumb accounts for arguably 40 percent of human hand use [1]. Thumb design is critical in all prosthetic hands and is more complex than the other fingers.

The Be-bionic 3 has an adjustable thumb which can be placed in an opposed or non-opposed position, the difference between these positions can be seen in the images below. This prosthesis cannot directly change the thumb's position. In order to switch between opposed and non-opposed positions the user must apply an external force to “click” the thumb into position, e.g. use the other hand to change the



Figure 2.7 – Opposed and Non-opposed thumb positions of the Be-bionic 3

Suppose a user of the Be-bionic 3 is using a computer mouse. If that user reaches to pick up a bottle of water not only, would they have to change the thumb position using their other hand, they would also need to cycle to a new grip state. The user would be better off using their other biological hand to fetch the water bottle in the first place. In such a situation this prosthesis provides no practical benefit. With all that being said, prosthetic devices are intended to provide more than functional practicality. For many amputees, the loss of an extremity is also accompanied by a significant decrease in confidence and self-esteem. A prosthetic arm can help alleviate these issues.

Physical appearance is an important aspect of prosthetic arms. A survey of myoelectric prosthetic hand users found that the majority of adult users were dissatisfied with their devices cosmetic appearance.

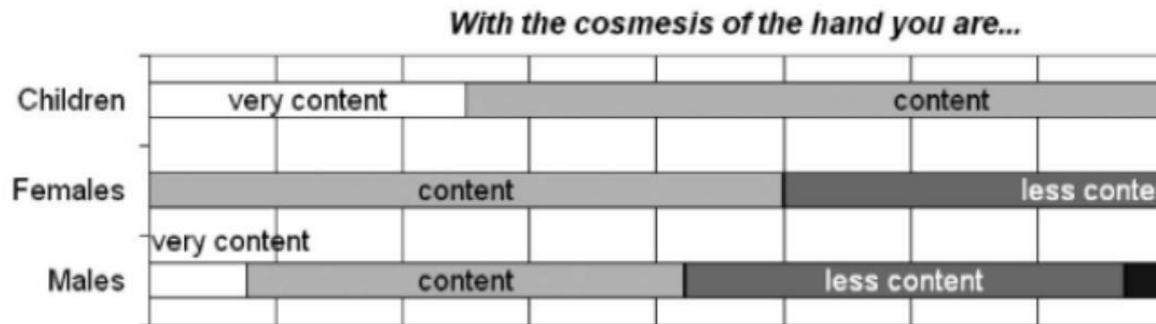


Figure 2.8 – Survey results of myoelectric prostheses users

The Be-bionic 3 is controlled through electro myography (EMG) electrodes placed on the surface of the user's skin. The placement of these electrodes depends on the level of amputation but is usually around the upper forearm. Be-balance computer software can be used to adjust a number of settings to enhance the user's control of the device and tune the system to the user's myoelectric signals

The level of EMG control is dependent on the specific amputation but is generally limited to only a few different commands which cannot be executed simultaneously.

In fact, it is due to this limitation in control that a predefined grip system and a manually adjustable thumb were designed. If a more complex system was designed the user would simply have no way of controlling iLimb digits

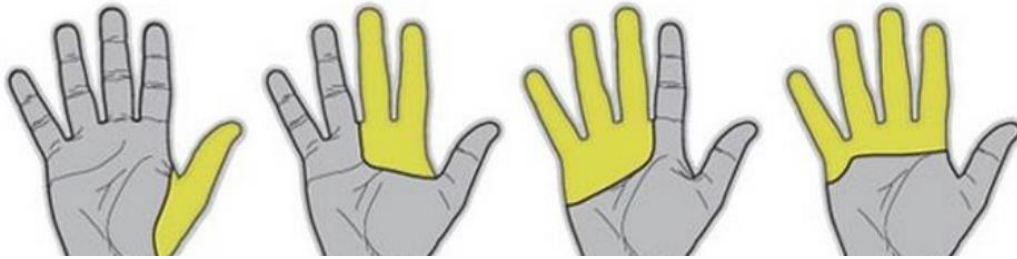
One major difficulty with developing prosthetic arms is that amputation can occur at any point along the arm and is unique in every case. The Be-bionic 3 arm previously discussed incorporates electric motors into the palm to actuate the fingers. As a result, the Be-bionic would be of no use to an amputee who has lost several fingers but still has their palm intact.

The iLimb digits developed by touch bionics incorporate electric motors directly into the prosthetic fingers. This allows for the palm area to fit into a socket connection attaching the prosthetic fingers to the hand. The image below shows possible amputations which would be suitable for use of the iLimb digits.



Figure

the



2.9 –  
Examples of  
amputations  
suitable for  
iLimb Digits

A custom socket is designed to fit around the remaining area of the user's palm and as many digits as necessary can be added to the system. Like the Be-bionic 3, the iLimb digits are controlled through EMG electrodes, which are placed over muscle regions in the palm. A small package must be worn around the user's wrist which contains the battery and controller for the system. A disadvantage of this system is that relatively small motors have to be used to be able to fit inside the fingers. This leads to digits which move slower and are weaker than those in other commercial prostheses.



Figure 2.10 – Limb Digits attached to an amputee's four fingers and the thumb but palm still intact



## **Problem Statement**

The loss or impairment of a limb significantly affects an individual's ability to perform everyday tasks, reducing their independence and quality of life. While prosthetic arms are available, many existing solutions suffer from limitations such as high costs, limited functionality, and lack of intuitive control, especially for complex hand movements. Furthermore, many prosthetics are heavy, uncomfortable, or difficult for users to adapt to due to insufficient sensory feedback.

This project aims to develop a low-cost, lightweight, and highly functional prosthetic arm that improves upon these limitations. The goal is to integrate advanced control systems, such as myoelectric sensors or brain-machine interfaces, to enable more natural and precise movements. Additionally, the design will prioritize user comfort and adaptability, allowing for ease of use in various daily activities.

By addressing these challenges, this project seeks to provide a more accessible and effective solution for amputees, enhancing their autonomy and ability to engage in regular physical activities.

## Objectives

### 1. Design and Development:

- To design and develop a functional prosthetic arm that mimics the natural range of motion of a human arm, focusing on the elbow, wrist, and hand movements.
- To create a lightweight and ergonomically comfortable prosthetic arm, suitable for long-term wear without causing user fatigue.

### 2. Cost Efficiency:

- To reduce the production cost of the prosthetic arm using affordable materials and manufacturing techniques, making it accessible to a larger population.
- To ensure that the prosthetic arm is easily repairable and has replaceable parts, extending its usability while minimizing long-term costs.

### 3. Control and Precision:

- To integrate an advanced control system (e.g., myoelectric sensors or EMG signals) that allows users to intuitively control the prosthetic arm with precise movements based on muscle signals.
- To ensure that the prosthetic arm responds quickly and accurately to user inputs, minimizing lag and enhancing control for fine motor skills (e.g., gripping small objects).

### 4. User Adaptability:

- To design a prosthetic that can be customized or adjusted based on the user's individual requirements (e.g., different hand sizes, levels of amputation).
- To develop a user-friendly interface or calibration system that helps users easily adapt to the prosthetic's functionality within a short learning period.

### 5. Sensory Feedback:

- To implement a basic sensory feedback system that provides the user with tactile or pressure information, improving their ability to interact with objects and surroundings more naturally.

## 6. Durability and Sustainability:

- To create a prosthetic arm that is durable and can withstand everyday wear and tear, maintaining functionality under various environmental conditions.
- To ensure the prosthetic is designed with sustainable materials where possible, reducing environmental impact.

## **Methodology**

### **1. Literature Review**

A comprehensive review of existing prosthetic technologies was conducted to understand current designs, control systems, and limitations. The review focused on:

- Myoelectric Types of prosthetic arms.
- Muscle sensors Control mechanisms
- Lightweight composites Materials used in prosthetics
- Current challenges in terms of functionality, cost, and user comfort.

### **2. Requirement Analysis**

Based on the literature review and interviews with potential users (amputees or clinicians), the following requirements were defined for the prosthetic arm:

- Control System: Capable of precise movement, responsive to user inputs.
- Weight and Material: Lightweight and durable, with materials that are comfortable to wear for extended periods.
- Cost: Affordable design, making it accessible to a larger population.
- Aesthetic Design: A visually acceptable prosthetic arm with human-like aesthetics.

### **3. Design and Simulation**

The design process involved the following steps:

- Conceptual Design: Initial sketches and computer-aided design (CAD) models were developed, focusing on the mechanical structure, joint movements, and overall appearance.
- Component Selection: Components such as motors, sensors (e.g., myoelectric sensors), microcontrollers, and batteries were chosen based on size, weight, and power consumption.
- Software Simulation: The mechanical parts and control system were simulated using and MATLAB/Simulink to ensure that the movements of the arm would be smooth and accurate.

#### **4. Prototype Development**

- **3D Printing & Assembly:** The structural components were 3D printed using lightweight materials, such as PLA or ABS, and then assembled. The actuation mechanisms (motors or pneumatic actuators) were integrated into the design.
- **Sensor Integration:** Myoelectric sensors were attached to the arm to capture muscle signals from the user's residual limb. These signals were used to control the movement of the prosthetic arm.
- **Control Algorithm Development:** A control algorithm was developed using a microcontroller (e.g., Arduino, Raspberry Pi) to process input from the sensors and actuate the motors for smooth, precise movements.

#### **5. Testing and Validation**

The prototype was tested in the following ways:

- **Functional Testing:** Basic functionality (grasping, rotating, and other hand movements) was tested in a controlled lab environment.
- **Usability Testing:** Feedback was gathered from a small group of users, including amputees, on the arm's comfort, ease of use, and responsiveness to their inputs.
- **Durability Testing:** The prosthetic arm was subjected to stress tests to evaluate the durability of its materials and components during various everyday tasks.
- **Battery Life:** Power consumption and battery performance were measured to ensure the arm could operate efficiently for a reasonable period before recharging.

#### **6. Iteration and Improvement**

Based on the testing phase, several iterations of the design were made to improve functionality, ergonomics, and responsiveness. Changes included:

- Adjustments to the sensor positioning and calibration.
- Refining the control algorithm for more precise and smoother movements.
- Re-designing certain mechanical components to reduce weight and improve aesthetics.

## **Specifications of the System**

### **1. Mechanical Specifications**

- Number of Degrees of Freedom (DoF):

The prosthetic arm offers 5 degrees of freedom for Thumb and 4 degrees of freedom for other Finger, allowing for movements

- Wrist rotation:  $\pm 90$  degrees.

- Elbow flexion:  $0^\circ$  to  $135^\circ$ .

- Grip Strength:

The maximum gripping force of the hand is 10 N to support daily tasks like holding objects.

### **2. Electrical Specifications**

- Power Source:

- Battery Type: Lithium-ion rechargeable battery.

- Battery Capacity: 2000 mAh

- Operating Time: 5 hrs of continuous usage before recharge is needed.

- Charging Time: Approximately 2 hours to fully charge.

- Motor Specifications:

- Type: DC servo motors.      - Voltage: 6V

- Torque: 3.5 Nm for each joint actuator, providing sufficient force for functional movements.

- Sensor Specifications:

- Myoelectric Sensors: Spartan EMG Sensors

Number of sensors: 3 sensors placed on the user's residual limb to capture muscle signals.

Sensitivity:  $\pm 2$  mV to detect even small muscle contractions.

- Feedback Sensors:

Position and force sensors in the hand for real-time feedback on grip strength and position accuracy.

Microcontroller/Processor:

- Type: Node MCU
- Processor Speed: 80 MHz to 160 MHz.
- Memory: 128 KB

### **3. Control System Specifications**

- Control Method:
- Myoelectric Control: The arm is controlled via muscle signals (EMG) from the user's residual limb.
- Signal Processing:
  - The EMG signals are processed using a microcontroller with an algorithm that translates these signals into movements (grip, rotation, flexion).
  - A control algorithm with real-time response (e.g., latency of less than 50 ms).
- User Interface:
  - Bluetooth connectivity allows users to adjust settings and monitor usage through a mobile app.
- Pre-programmed Movements:
  - The arm can switch between different modes (e.g., power grip, precision grip) based on muscle signals or manual mode switching.

### **4. Functional Specifications**

- Movement Range:
  - Elbow Flexion/Extension: [Specify range, e.g., 0° to 135°]
  - Wrist Rotation: [Specify, e.g., ±90°]
- Finger Movements: Individual finger control for complex tasks or grouped finger control for simpler tasks.

- Grip Types Supported:
- Power Grip: For holding larger objects like a bottle.
- Precision Grip: For holding smaller objects like a pen.
- Lateral Pinch: For pinching thin objects like paper.
- Speed of Movement:
- Maximum speed of hand opening/closing: [Specify speed, e.g., 0.5 seconds for full open/close]\_.
- Wrist rotation speed: \_[Specify speed, e.g., 10 degrees per second]\_.

## **5. Safety Features**

- Overload Protection:  
Automatic shutdown of motors in case of excessive force detected in the joints to prevent damage.
- Emergency Stop:  
The system includes an emergency stop function that allows the user to instantly halt all movements in case of malfunction or danger.
- Battery Management System (BMS):  
Integrated BMS to prevent overcharging and overheating of the battery.

## **6. Aesthetic and Comfort Specifications**

- Outer Shell Material:  
Silicone skin-like material to give the prosthetic arm a more natural appearance.
- Comfort Features:
- Soft inner lining in the socket to prevent irritation and discomfort.
- Adjustable straps and fittings for custom sizing to fit different residual limb shapes.
- Weight Distribution:  
Balanced design to ensure even weight distribution and reduce strain on the user's shoulder.



## **7. Cost and Availability**

- Estimated Production Cost:

The total estimated production cost is [e.g., \$500] per unit, designed to be affordable while maintaining functionality.

## Block Diagram

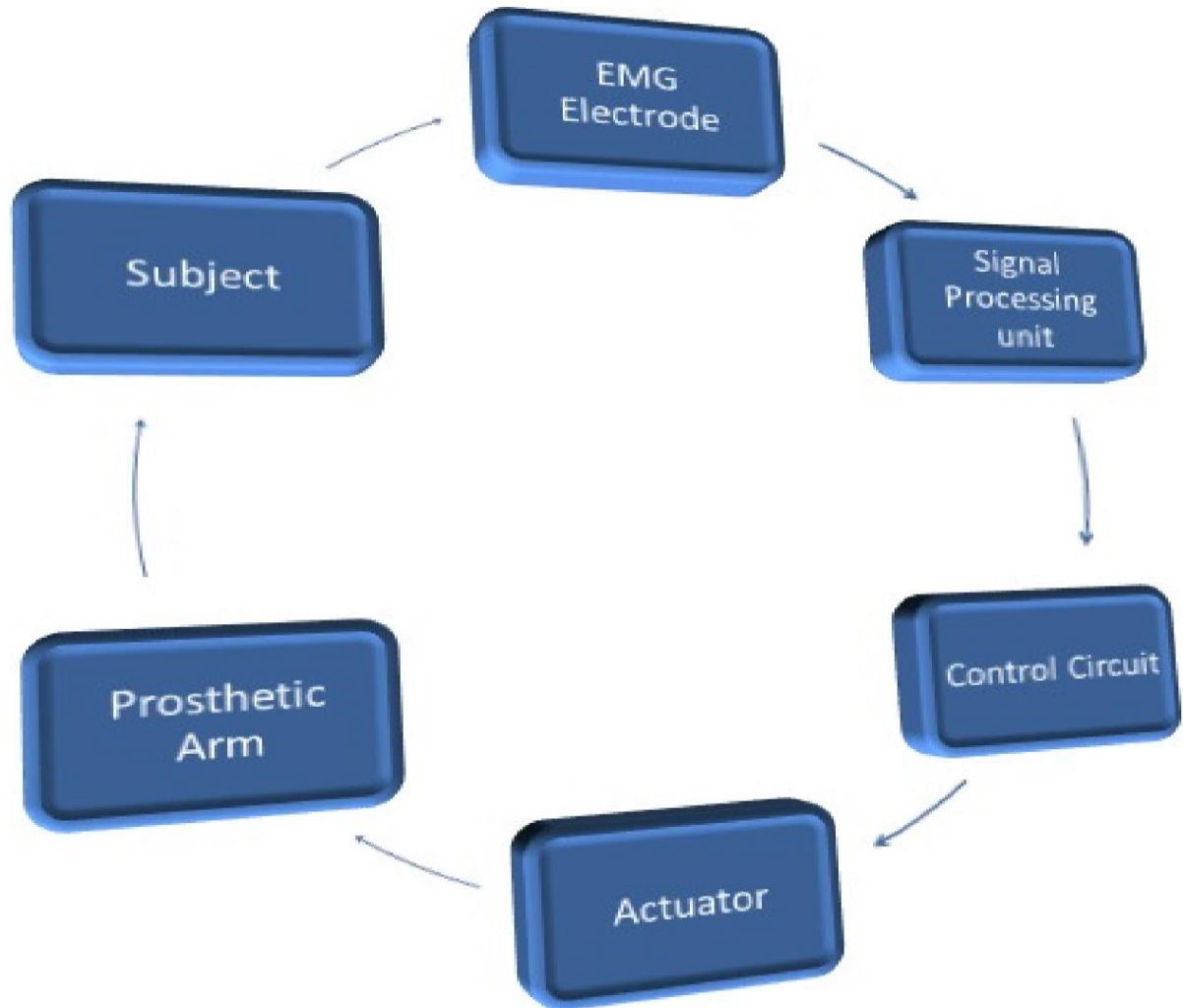


Figure 2.11 – Block Diagram

The block diagram illustrates the functional flow of a myoelectric prosthetic arm system, which enables an amputee to control a prosthetic arm using their own muscle signals. The process begins with the subject, the user of the prosthetic arm, who still has residual muscle tissue in the limb. When the user attempts to move the missing limb, the muscles generate electrical signals, which are detected by EMG electrodes placed on the skin. These electrodes pick up the weak electrical signals, known as electromyographic (EMG) signals, from muscle contractions.

The signal processing unit then amplifies, filters, and processes these EMG signals to extract meaningful data, such as the strength and pattern of the muscle contractions. This processed data is sent to the control circuit, which interprets the information and converts it into control commands for the prosthetic arm. The control circuit directs the actuators, which are motors or other mechanical components, to perform specific movements, such as rotating the wrist, opening the hand, or flexing the elbow

Finally, the movements are carried out by the prosthetic arm, which is the physical device worn by the user. The prosthetic arm mimics the natural movements of a human limb, allowing the user to grasp objects, manipulate items, and perform everyday tasks. This entire process operates in a feedback loop, where the prosthetic arm's movements may provide feedback to the user, enabling them to fine-tune their muscle signals for more precise control. In this way, the system allows for intuitive control of the prosthetic arm, enhancing the user's ability to perform functional movements with a high degree of independence.

## Algorithm

### 1. Main Loop Execution:

- Step 1: Myoelectric Input

The system receives a myoelectric input through the contraction of muscles. This signal is captured using Electromyography (EMG) sensors.

- Step 2: Analog-to-Digital Conversion (ADC) of EMG Signals

The captured EMG signals are converted from analog to digital format for further processing.

- Step 3: Gather Sensor Data and Control Signals

The system gathers data from various sensors and generates control signals for servo motors and vibrotactile feedback motors.

- Step 4: Analog-to-Digital Conversion (ADC) of Pressure Sensor Signals

Pressure sensor data is also converted from analog to digital format to provide sensory information on the grip or pressure applied.

- Step 5: Feedback and User Adjustment

The processed information is used to provide vibrotactile sensory feedback to the user. If necessary, the user can make adjustments based on the feedback.

## Flowchart:

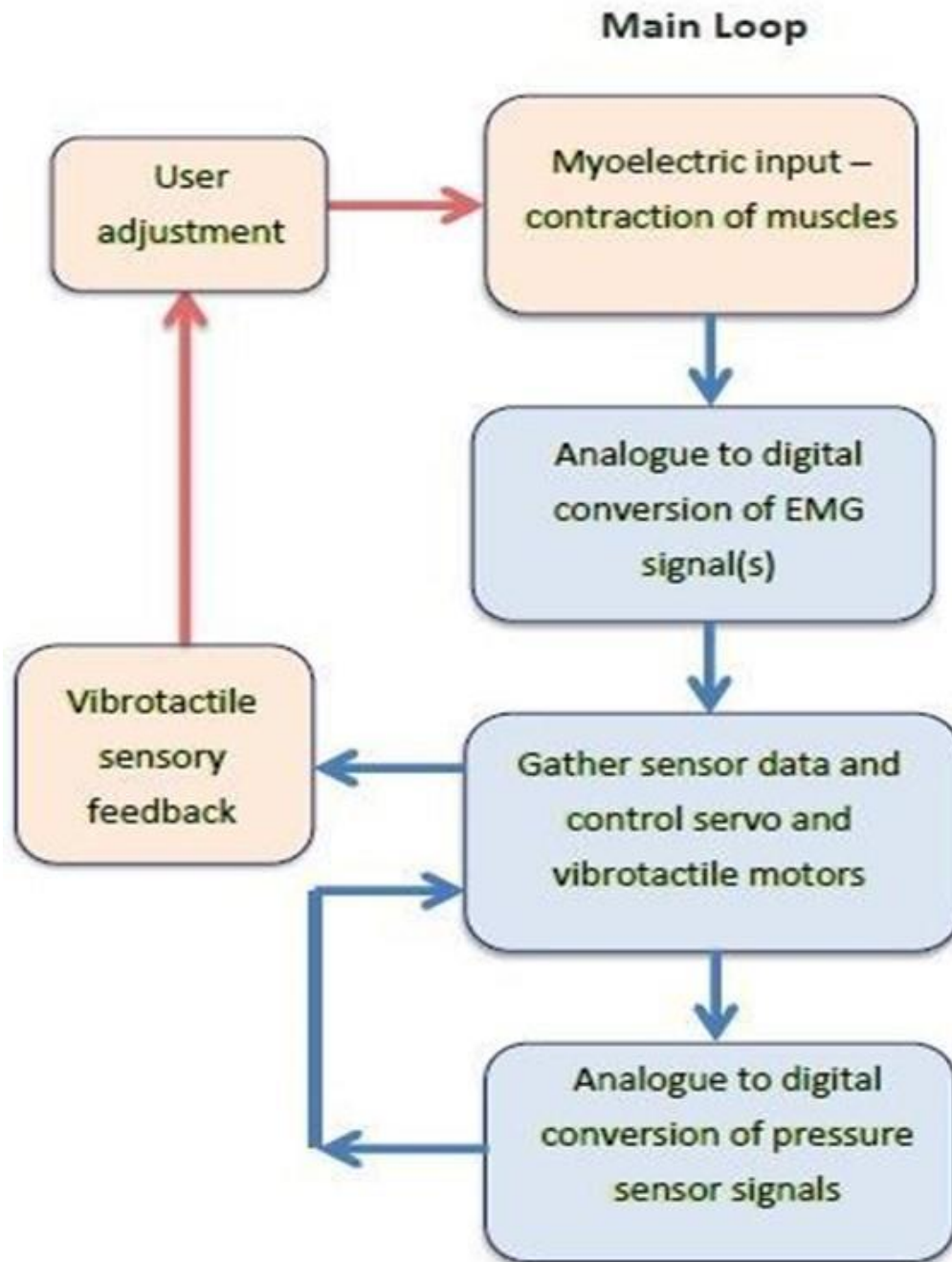


Figure 2.12- Flowchart

## Summary

This project focuses on the design and development of an affordable, lightweight, and functional prosthetic arm aimed at improving the quality of life for amputees. The prosthetic arm was designed to address the limitations of existing solutions, including high cost, limited functionality, and user discomfort. Through the integration of advanced control systems, such as myoelectric sensors, the arm provides users with precise and intuitive control over various hand and arm movements.

The project began with a detailed analysis of current prosthetic technologies and user requirements, leading to the selection of materials and components that balance performance, weight, and cost. A combination of 3D-printed components and off-the-shelf actuators enabled the creation of a functional prototype that supports multiple degrees of freedom, such as wrist rotation, elbow flexion, and a variety of hand grips. The control system utilizes myoelectric sensors to capture muscle signals from the user's residual limb, which are processed by a microcontroller to generate smooth, natural movements.

The prototype was tested for functionality, comfort, and durability. The results showed that the arm performs well in tasks such as gripping objects, rotating the wrist, and flexing the elbow, with a responsive and user-friendly control system. Feedback from test users, including amputees, highlighted the comfort of the prosthetic and its ease of use, with further iterations made to optimize performance.

In conclusion, the prosthetic arm developed in this project offers a promising solution for amputees seeking a functional, cost-effective, and comfortable device. It demonstrates the potential for future improvements and broader adoption in the field of prosthetics, ultimately contributing to enhancing the autonomy and quality of life for users.

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